

AI Transformation Portal

Architecture & Data Flow — Technical Documentation

المنصة الحكومية لتخطيط ومتابعة مشروع الذكاء الاصطناعي المساعد

Document 1 of 2 · Prepared for the IT team (staging deployment, DB linking, go-live)

Version 2.0 · July 2026 · Companion documents: docs/it-handover.md, .env.example, Roles & Access (PDF)

1. System overview

The portal is a **Next.js 14** application (App Router, TypeScript, RTL Arabic UI) backed by **PostgreSQL 16** through **Prisma ORM**. Users sign in with **UAE PASS** (OIDC); sessions are stateless HMAC-signed httpOnly cookies (no sessions table). An optional internal, OpenAI-compatible model powers the AI review step. One codebase serves all the roles — **مشرف النظام** (admin), **منسق المسار في الجهة, رئيس**, **اللجنة الوطنية (الرئيس والأمانة العامة)**, and **المسار ونائبه**; every screen derives from a single role-scoped view-model, so what a user can see is decided in one place. The **admin** gets a dedicated console for managing users & roles and assigning the stream heads and national committee.

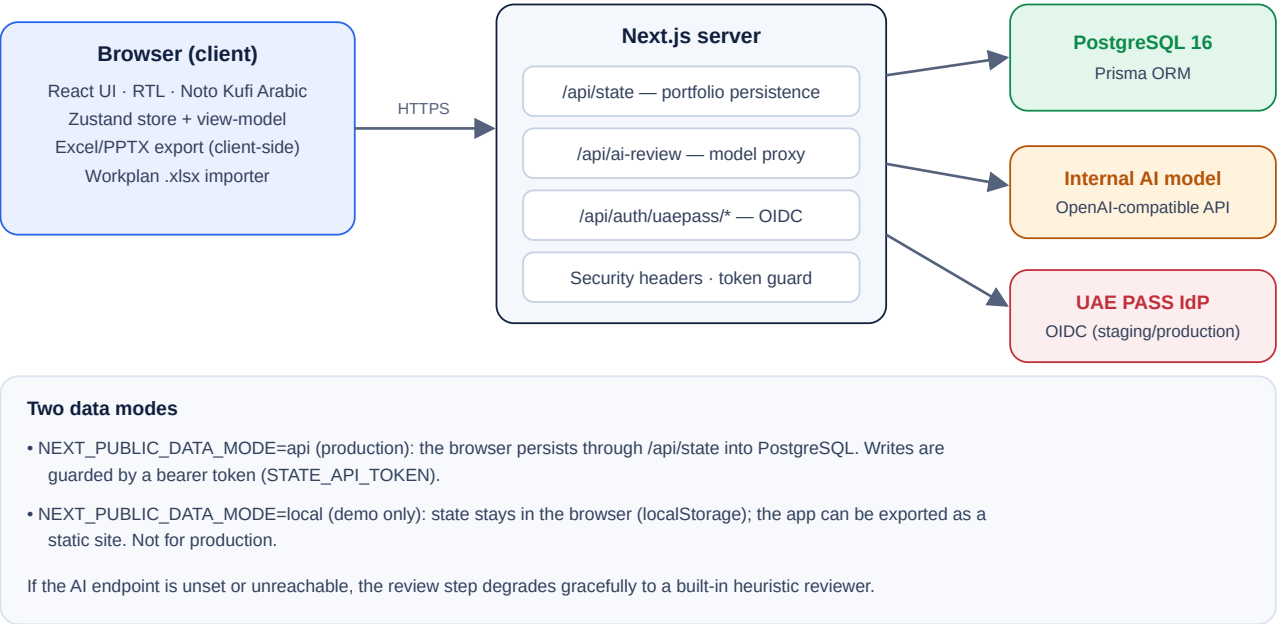


Fig. 1 — Runtime architecture

2. Deployment architecture (Kubernetes)

Manifests are provided in **k8s/**. The application image is built from the repository **Dockerfile**; database schema changes run as a one-shot migration **Job** before each rollout.

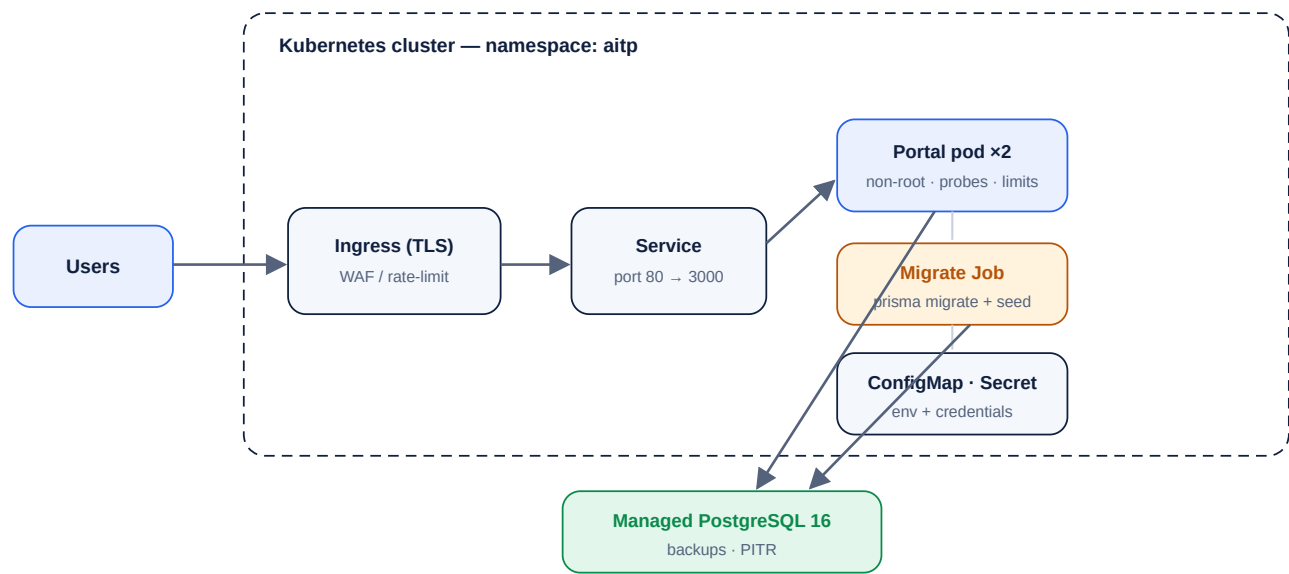


Fig. 2 — Kubernetes deployment (k8s/ manifests)

Release flow

#	Step	Command / artifact
1	Build & push image	<code>docker build -t REGISTRY/aitp-portal:TAG .</code>
2	Apply namespace, config, secret	<code>k8s/namespace.yaml · configmap.yaml · secret (from secret.example.yaml)</code>
3	Migrate + seed reference data	<code>k8s/migrate-job.yaml (idempotent)</code>
4	Roll application	<code>k8s/deployment.yaml · service.yaml · ingress.yaml</code>

3. Data model

Schema source of truth: **prisma/schema.prisma** + migrations **0001-0010** (verified on a fresh PostgreSQL 16). The seed loads **reference data + roles + starter accounts**; items and all transaction tables start empty (the demo portfolio only seeds when NEXT_PUBLIC_DEMO_DATA=1).

Group	Tables	Notes
Reference	streams, entities, program_phases, exec_batches, service_catalog, settings	3 streams (ops · strategy · services) in official order with the predefined heads (رؤساء دفعات الإطلاق / التقييم والتهيئة، الدفعات الأولى...السادسة; الفيدرالية); federal entities; دليل الخدمات (migration 0010) holds entity-scoped الخدمة/الخدمة الفرعية — الاتحادية pairs feeding the services-stream dropdowns.
Access control	roles, permissions, role_permissions, users, user_roles, user_entity_scopes, user_stream_scopes	roles holds the backend role codes with the seeded permission matrix (role_permissions); users.role is a FK into it. Seed creates starter accounts (admin, committee, 3 stream heads, one coordinator per stream) keyed by @aigp.gov.ae emails IT re-points to UAE PASS identities. Sessions are stateless signed cookies.
Team records	stream_owners, entity_reps (legacy)	Per-stream owners inside each entity. All accounts are provisioned centrally by the system admin — there is no self-service team setup.
Portfolio	items, item_launch_plans, launch_plans, launches, item_launches, exec_checklist_items, sub_milestones	Items are مهمة / خدمة with the per-stream form fields, the دفعة assignment (exec_batch) and the returned-with-notes state. Statuses: مسودة / قيد الاعتماد / للتعديل (locked). launch_plans/launches are legacy tables kept for compatibility.
Workflow / audit	log_entries, nominations, fundings, funding_cancellations, notifications	Every action is logged with actor + timestamp. nominations / fundings are legacy (the concepts are out of the current flow). notifications persist الإشعارات (per-user or role/entity/stream broadcast) for the mandatory email push via POST /api/notify.
Misc	app_state	Serialized UI-state snapshot used by /api/state.

Fig. 3 — Data model groups

4. Data flow — sign-in (UAE PASS OIDC)

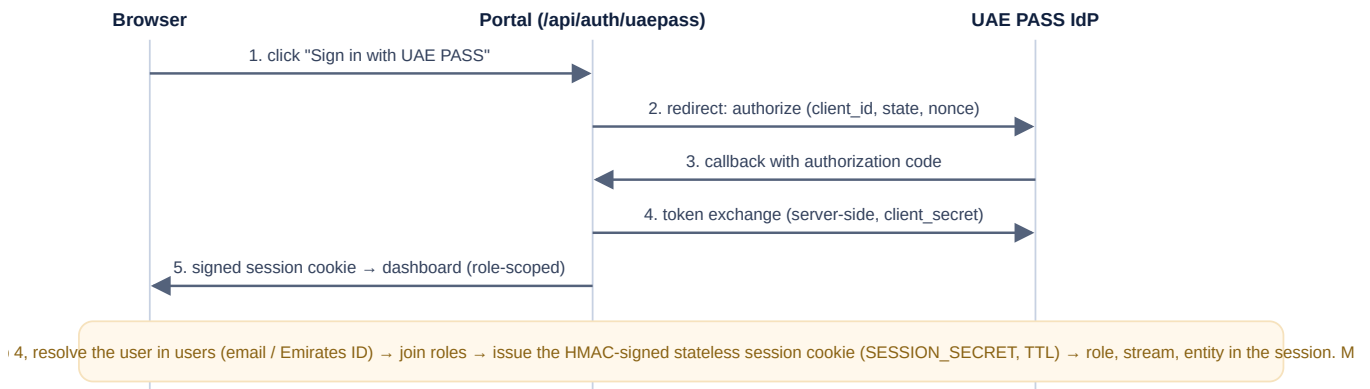
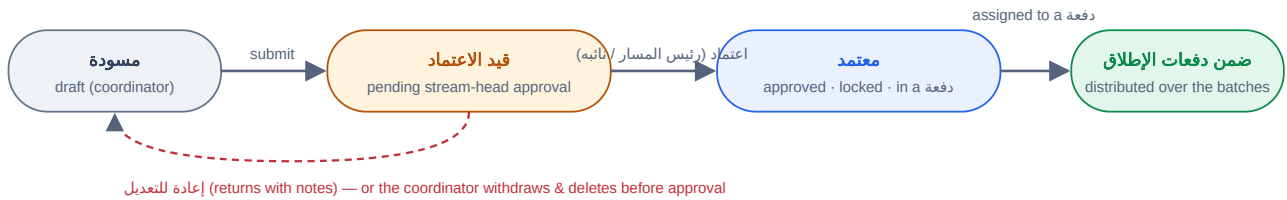


Fig. 4 — Authentication sequence

5. Data flow — item lifecycle



دفعات الإطلاق (after approval)

معتمد entries are distributed over their stream's (الأولى...السادسة) دفعات; start/end dates are bounded to each دفعة window.
Moving an entry between دفعات notifies رئيس المسار; entries can never sit on another stream's batches.
Every transition writes a log entry (actor, action, timestamp) — the item's audit trail in the detail page.

Fig. 5 — Lifecycle state machine + دفعات الإطلاق

6. Data flow — bulk import (workplan file)

The coordinator downloads the **per-stream Excel template** from the bulk-upload step («تنزيل نموذج المسار») — its columns are generated from the same field definitions as the entry form (STREAM_FIELDS), including dropdowns for every select field. In the services stream the الخدمة الفرعية / الخدمة columns are dropdowns from the entity's own service_catalog rows.

Step	What happens
1 · Upload .xlsx	The importer finds the header row by matching the exact column labels first (fuzzy match only as a fallback), one column per field.
2 · Normalize & derive	Rows are normalized like the form: conditional fields are cleared when inapplicable (نوع عملية الدعم, نظام/نسبة الأتمتة), automation is clamped ($95 \geq$ جزئياً = 100%, كلياً = 100%), and أولوية التحول is derived from the stream matrix.
3 · Review screen	Every row shows its verdict; rows with missing required fields are flagged «بيانات ناقصة». A confirm button commits the import.
4 · Commit	Complete rows are submitted (رئيس المسار — قيد الاعتماد is notified); incomplete rows are saved as drafts. Selecting drafts later offers «إكمال وإرسال» / «إرسال للاعتماد» with a popup listing the missing fields.

Nothing is invented: whatever the file does not carry stays blank for manual completion.

8. Access provisioning & the admin console

Accounts live in **users** (FK **users.role** → **roles.key**). **One actor provisions everyone:** the **system admin** creates and manages all accounts for all entities — stream heads and deputies (with their stream), coordinators (with entity + stream), and the national committee. There is no self-service team-setup screen.

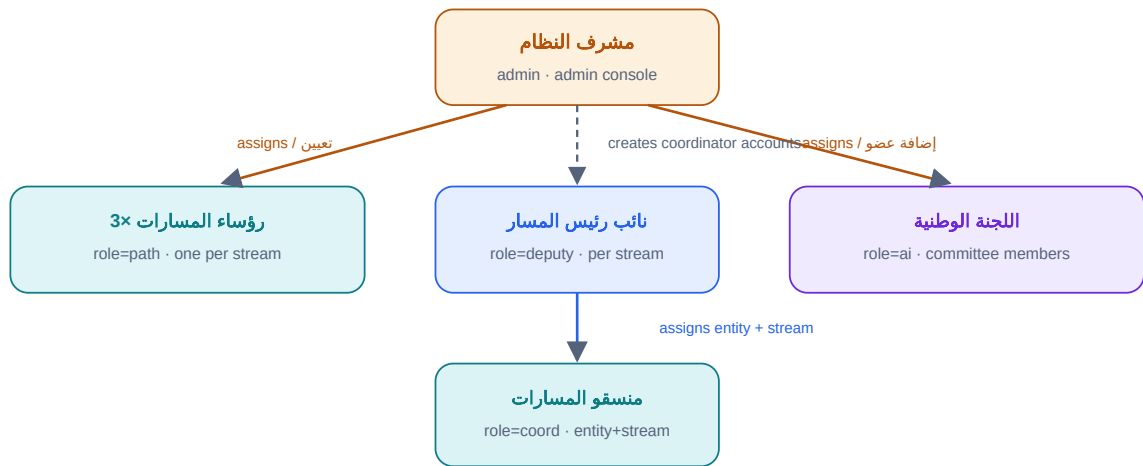


Fig. 7 — Who provisions whom (admin console vs. entity team setup)

Roles & capabilities (roles.permissions)

role	الدور	scope	Key capabilities
admin	مشرف النظام	system	users.manage · roles.view · streamhead.assign · committee.assign
ai	اللجنة الوطنية	national	item.view.all (approved only) · phase.edit · ai_review.run
path	رئيس المسار	stream	item.view.stream · item.approve · item.return
deputy	نائب رئيس المسار	stream	item.view.stream · item.approve · item.return
coord	منسق المسار في الجهة	entity+stream	item.create · item.update · item.submit · plan.edit

Provisioning rules enforced by the app: **coord** and **entity** require `entity_id`; **coord** and **path** require `stream_id`; **ai** and **admin** require neither. Deactivate (never delete) accounts via `is_active=false` so the audit trail keeps valid author names.